

NLP-Based Sentiment Analysis System Using Naïve Bayes and Transformers

Problem Statement and Literature Survey

1. Problem Statement

Sentiment analysis is a major natural language processing task in which user-generated text such as reviews, comments and posts is examined to determine the emotional tone expressed by its author, typically classified as positive, negative, or occasionally neutral. Traditional dictionary-based, rule-based and manual annotation approaches are limited against informal, colloquial, slang-heavy or sarcastic text that does not appear in standard lexicons. Machine-learning and deep-learning techniques can learn patterns from labeled positive and negative text and automate sentiment classification, while transformer-based models can capture contextual meaning beyond individual words.

Nevertheless, existing AI-based sentiment analysis systems face important challenges including dataset dependency, weak generalisation to unseen text distributions or domains, informal and symbolic language, implicit or mixed sentiment within a single sentence, class imbalance, false positives, computational cost and limited explainability of model decisions. The three selected studies address these issues from complementary perspectives: hybrid dictionary–machine-learning classification for short informal text, comparative evaluation of deep networks and large language models for aspect-based sentiment, and topic-model-driven sentiment analysis of social media discussions.

Therefore, the proposed project aims to develop an AI-powered NLP-based sentiment analysis system that extracts relevant textual features, trains and compares a classical Naïve Bayes classifier and a transformer-based classifier, evaluates them using multiple metrics and emphasises generalisation to unseen data. The system should provide a clear positive/negative decision together with a useful sentiment probability and feature-level explanation.

2. Literature Survey

1. Sentiment Analysis of Danmaku Videos Based on Naïve Bayes and Sentiment Dictionary

Author(s): Zhi Li; Rui Li; Guanghao Jin (2020)

Algorithm /Methodology: Hybrid dictionary–machine-learning system (SD-NB). A danmaku sentiment dictionary is constructed from the Dalian University of Technology emotional

vocabulary ontology (27,476 words across seven emotion dimensions), a Catchwords dictionary of 733 network hot words, and a manually weighted set of 161 emoticons. Dictionary-based feature extraction replaces n-grams, and a Naïve Bayes classifier with Laplace smoothing performs polarity classification. A nine-case weight-calculation scheme handles sentiment words, network words and emoticons modified by negative words and degree adverbs. Dataset: 541,841 danmaku reviews scraped from Tencent Video (reduced to 301,052 after preprocessing; 65,000 manually labeled for evaluation).

Advantages: Combines complementary lexicon and machine-learning approaches; dictionary side requires no large labeled corpus and is fast and interpretable; catchwords and emoticons handle short, informal and symbolic text that standard lexicons miss; avoids semantic confusion caused by fixed-length n-gram windows; reports accuracy, precision, recall and F1 for both classes; achieved 88.2% accuracy, outperforming N-gram-NB (75.0%), N-gram-SVM (73.5%) and TextCNN (76.8%); produces dynamic time-distribution visualisations of seven sentiment dimensions.

Disadvantages /Limitations: Manual construction and upkeep of the catchword and emoticon lexicon does not scale to new slang. Evaluation is centred on binary polarity despite the seven-dimension design, and on a single platform and genre (Chinese variety-show danmaku), so independent cross-dataset generalisation is not demonstrated. The rule-based treatment of negation and degree adverbs cannot handle sarcasm or implicit sentiment. Relevance of emoticon weights is platform-specific.

2. Comparative Analysis of Deep Natural Networks and Large Language Models for Aspect-Based Sentiment Analysis

Author(s): Nimra Mughal; Ghulam Mujtaba; Sarang Shaikh; Aveenash Kumar; Sher Muhammad Daudpota (2024)

Algorithm /Methodology: Systematic benchmarking of three model generations for aspect-based sentiment analysis (ABSA): LSTM-based models (AE-LSTM, AT-LSTM, ATAE-LSTM) trained from scratch; fine-tuned transformer ABSA models (DeBERTa-v3-base-ABSA-v1.1, flan-t5-large-ABSA); and zero-shot large language models (PaLM text-bison-001, GPT-3.5-Turbo) driven by structured prompt engineering. Two subtasks are evaluated (ATSA and ACSA) on five benchmark datasets (SemEval-2016, Sentihood, MAMS, YASO, DOTSA) under in-domain and cross-domain settings using overall accuracy and weighted F-measure.

Advantages: First comprehensive cross-model, cross-domain comparison including zero-shot LLMs; directly quantifies domain sensitivity and cross-dataset degradation of trained models; demonstrates that LLMs can deliver strong results without any task-specific training or labeled data (PaLM 93.5% on SemEval16/Restaurant and 98.1% on YASO, beating fine-tuned DeBERTa and published baselines); shows fine-tuned DeBERTa excels in its training domains (93.1% on DOTSA/Hotels); clarifies when fine-tuning versus prompting is preferable.

Disadvantages /Limitations: LLMs perform poorly on implicit, multi-aspect data (PaLM only 48.8% ATSA and 55.8% ACSA on MAMS, well below DeBERTa's 77.4–83.8%), showing zero-shot prompting does not solve mixed-sentiment sentences; prompt sensitivity and robustness are not quantified; only older LLMs (GPT-3.5, PaLM) are tested, and API usage introduces cost and latency; trained-model results remain tied to the benchmark datasets used; no error analysis at the linguistic level.

3. Exploring Public Converse on AI via Reddit: A Topic Modeling and Sentiment Analysis Approach

Author(s): Xi Chen; Tala Talaei Khoei; Aditi Singh (2026)

Algorithm /Methodology: Mixed topic-modeling and sentiment pipeline. 29,879 pseudo-documents (posts plus root comments) are collected from 41 AI-related subreddits via PRAW. Three topic models (LDA, NMF, BERTopic) are compared using OCTIS with topic diversity, C_v coherence and KL-uniform metrics. Topic representations are fine-tuned through a semi-AI approach in which GPT-4o generates topic descriptions that are manually restructured into a four-level taxonomy (primary, principal, tertiary fields plus description). Analysis combines Sankey diagrams, dynamic topic modeling and lexicon-based compound sentiment scoring of topics and fields.

Advantages: Broadens AI-discourse research beyond single products such as ChatGPT to a multi-subreddit, multi-domain corpus; semi-AI topic tuning improves interpretability while reducing subjective labeling bias; BERTopic achieved the best topic quality (diversity 0.69, coherence 0.58, KL-uniform 4.04), exceeding coherence scores reported in comparable studies; analysis is genuinely high-dimensional, linking topic composition, temporal dynamics and sentiment; derives actionable findings, e.g., that task complexity and access barriers depress sentiment while customization raises it.

Disadvantages /Limitations: Only root comments are collected, so conversation depth and threaded replies are lost; sampling per subreddit is uneven (API caps), biasing topic frequencies; lexicon-based compound sentiment cannot capture sarcasm, irony or implicit opinion; trends over time are described qualitatively without statistical hypothesis testing; taxonomy construction remains partly manual and thus partially subjective; user-privacy aggregation limits behavioral-level conclusions.

3. Comparative Literature Survey

Paper	Algorithm/Methodology	Advantages	Disadvantages/Limitations
Sentiment Analysis of Danmaku Videos	Hybrid SD-NB. Sentiment dictionary (DUT ontology + 733 catchwords + 161	No large labeled corpus needed on the lexicon side; fast and interpretable; handles informal/symbolic	Manual lexicon upkeep does not scale; binary-only evaluation despite seven dimensions; single

Based on Naïve Bayes and Sentiment Dictionary	weighted emoticons) for feature extraction; Naïve Bayes with Laplace smoothing for classification; nine-case weight rules for negation and degree adverbs. 541,841 danmaku reviews; 65,000 manually labeled.	text; avoids n-gram semantic confusion; 88.2% accuracy, outperforming N-gram-NB, N-gram-SVM and TextCNN; dynamic seven-dimension sentiment visualisation.	platform and genre; no independent cross-dataset test; rules cannot handle sarcasm or implicit sentiment; platform-specific emoticon weights.
Comparative Analysis of Deep Natural Networks and Large Language Models for Aspect-Based Sentiment Analysis	Benchmarks LSTM family (AE/AT/ATAE-LSTM), fine-tuned transformers (DeBERTa-v3-ABSA, flan-t5-large-ABSA) and zero-shot LLMs (PaLM, GPT-3.5-Turbo) with prompt engineering on five datasets (SemEval16, Sentihood, MAMS, YASO, DOTSA); in-domain and cross-domain evaluation by accuracy and weighted F1.	First systematic LLM-versus-ABSA comparison; directly quantifies domain sensitivity and cross-dataset degradation; zero-shot PaLM reaches 93.5–98.1% on some datasets without any training data; shows when fine-tuning beats prompting; standardised multi-dataset protocol.	LLMs fail on implicit multi-aspect data (MAMS: PaLM 48.8% ATSA, 55.8% ACSA vs DeBERTa 77.4–83.8%); prompt sensitivity unquantified; only GPT-3.5/PaLM tested; API cost and latency; trained models remain dataset-bound; no linguistic error analysis.
Exploring Public Converse on AI via Reddit: A Topic Modeling and Sentiment Analysis Approach	29,879 pseudo-documents from 41 subreddits via PRAW; LDA, NMF and BERTopic compared with OCTIS (topic diversity, C_v coherence, KL-uniform); GPT-4o semi-AI topic fine-tuning into a four-level taxonomy; Sankey diagrams, dynamic topic modeling and lexicon-based compound sentiment analysis.	Multi-subreddit scope beyond single AI products; semi-AI labeling reduces subjective bias; BERTopic best on all metrics (diversity 0.69, coherence 0.58) and above comparable studies; high-dimensional composition/time/sentiment analysis; actionable findings on complexity, barriers and customization.	Root comments only (no threaded depth); uneven per-subreddit sampling; lexicon sentiment misses sarcasm and implicit opinion; no statistical testing of trends; taxonomy partly manual; privacy aggregation limits behavioral conclusions.

4. Research Gap Identified

- Dataset dependency: high within-dataset accuracy (e.g., 88.2% for SD-NB) may not transfer to independently collected text from other platforms or domains.
- Generalisation: sentiment expression changes continuously with slang, domains and platforms, and previously unseen text patterns remain difficult for all model families.
- Implicit and mixed sentiment: sentences containing contrasting opinions toward different aspects remain hard for lexicons, trained transformers and even zero-shot LLMs (MAMS results).
- Informal and symbolic language: emoticons, catchwords and misspellings require domain-specific resources whose manual maintenance does not scale.
- Label cost versus performance: fine-tuned transformers need substantial labeled data and compute, while cheap classical models sacrifice contextual accuracy.
- Explainability: practical users need understandable reasons behind a positive/negative decision, such as the influential words identified by dictionary-based or feature-importance methods.
- Evaluation practice: many studies report only within-dataset, often binary, results; cross-dataset validation and fine-grained (multi-class or aspect-level) evaluation remain uncommon.

5. References

Z. Li, R. Li, and G. Jin, "Sentiment Analysis of Danmaku Videos Based on Naïve Bayes and Sentiment Dictionary," *IEEE Access*, vol. 8, pp. 75073–75084, 2020. DOI: 10.1109/ACCESS.2020.2991738.

N. Mughal, G. Mujtaba, S. Shaikh, A. Kumar, and S. M. Daudpota, "Comparative Analysis of Deep Natural Networks and Large Language Models for Aspect-Based Sentiment Analysis," *IEEE Access*, vol. 12, pp. 60943–60960, 2024. DOI: 10.1109/ACCESS.2024.3384912.

X. Chen, T. T. Khoei, and A. Singh, "Exploring Public Converse on AI via Reddit: A Topic Modeling and Sentiment Analysis Approach," *Journal of Social Computing*, vol. 7, no. 1, pp. 94–109, March 2026.

Web sources used for verification: IEEE Xplore records for the two IEEE Access papers and the Journal of Social Computing record for the third paper. The analysis above distinguishes reported methodology/results from limitations inferred from the scope and evaluation design.